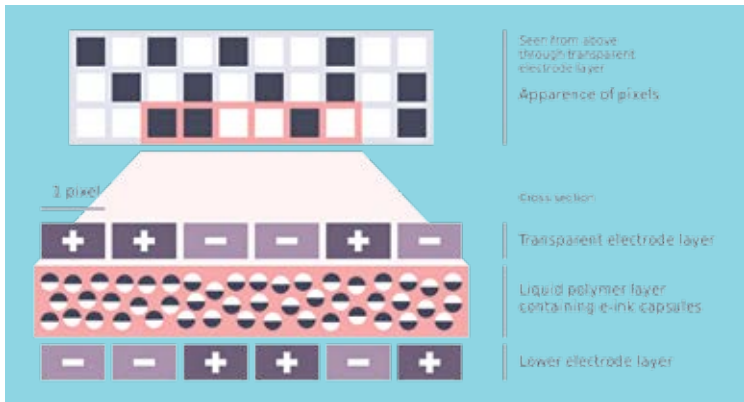


a time when it was not around. Currently, e-Paper displays offer a significantly low power consumption along with really wide viewing angles which makes it the perfect choice for ebook readers, low power electronic devices, and even some wearable devices. All of this is achievable due to the underlying technology.

How does an ePaper display work?

The main technical difference between an e-Paper display and say, a conventional LCD is that the LCD works on light generated by it, whereas an e-Paper display works on light reflected by it. The first ePaper display, known as Gyricon, was developed at Xerox in the '70s and consisted of polyethylene spheres that were Janus Particles with each half having the opposite charge as well as opposite colours (black/white) to the other. The spheres were suspended in a bubble of oil within a transparent silicon



The basic functionality of an e-Ink display with the sample output shown.

sheet to ensure their free rotation. Based on the voltage applied, the spheres would rotate accordingly with the white or the black side facing upwards.

This technology was further improved by the advent of electrophoretic displays that worked in a different method. A hydrocarbon oil is used to disperse titanium oxide particles along with a dark coloured dye, surface tension lowering agents and charging agents. Put in between two parallel conducting plates, separated by tens of micrometers. On application of voltage between the two plates, the particles swarm to the plate with the